

Sudarshan Babu

Research Scientist – Generative Modeling

sudarshan@ttic.edu
people.cs.uchicago.edu/~sudarshan

Research Summary

Research scientist specializing in diffusion and score-based generative models for image, video, and 3D scene generation, extending to cryo-EM volumes, 3D molecular structures, and single-cell data. Builds data-efficient generative systems with diffusion, flow matching, hypernetworks, and implicit neural representations, with emphasis on out-of-distribution generalization.

Core Expertise

Techniques	Diffusion & score-based modeling, flow matching, representation learning, meta-learning
Architectures	Hypernetworks, implicit neural representations (INRs), transformers
Modalities	Image, video, 3D scenes, cryo-EM volumes, 3D molecular structures, single-cell data

Professional Experience

Sep 2025–Present	Research Scientist , <i>Chan Zuckerberg Biohub Chicago</i> , Chicago, IL Developed scLDM, a latent diffusion model for single-cell data generation Built implicit neural field model for cryo-ET volume generation
Jan 2025–Sep 2025	AI Fellow , <i>Chan Zuckerberg Biohub Chicago</i> , Chicago, IL Developed HyperDiffusionFields (HyDiF), combining diffusion models with hypernetworks for implicit molecular neural fields generation
Oct 2021–Jan 2022	Research Scientist Intern , <i>Amazon</i> , Bellevue, WA Designed curriculum learning strategies for efficient neural network training
Feb 2021–Jun 2021	Research Scientist Intern , <i>NVIDIA</i> , Remote Built systems for training models on non-stationary long-tail distributions Contributed to 4 patent applications in adaptive learning and distributed training systems

Education

Oct 2019–2025	Ph.D., Computer Science , <i>Toyota Technological Institute at University of Chicago</i> , Chicago, IL, GPA: 3.90/4.0 Advisors: Prof. Michael Maire, Prof. Greg Shakhnarovich Dissertation: Acquiring and Adapting Priors for Novel Tasks via Neural Meta-Architectures
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Publications

Under submission	Atelier: Learning Local Self-Supervised Features for CryoEM Volumes via Hypernetworks Phillip Lo*, Sudarshan Babu*, Dari Kimanius [†] , Aly A. Khan [†]
ICML '26	Virtual Cells Need Context, Not Just Scale Payam Dibaieinia, Sudarshan Babu, Mei Knudson, Ali ElSheikh, et al.
ICML '26	A Scalable Latent Diffusion Model for Single-Cell Gene Expression Data Giovanni Palla, Sudarshan Babu, Payam Dibaieinia, et al.
NeurIPS WS '25	HyperDiffusionFields (HyDiF): Diffusion-Guided Hypernetworks for Learning Implicit Molecular Neural Fields Sudarshan Babu*, Phillip Lo*, Xiao Zhang, et al.
ICML '24	HyperFields: Towards Zero-Shot Generation of NeRFs from Text Sudarshan Babu*, Richard Liu*, Avery Zhou*, et al.
SIGGRAPH Asia '24	iSeg: Interactive 3D Segmentation via Interactive Attention Itai Lang*, Fei Xu*, Dale Decatur, Sudarshan Babu, Rana Hanocka
NeurIPS '21	Online Meta-Learning via Learning with Layer-Distributed Memory Sudarshan Babu, Pedro Savarese, Michael Maire
CVPR '21	Domain-Independent Dominance of Adaptive Methods Pedro Savarese, David McAllester, Sudarshan Babu, Michael Maire
Preprint '20	HyperNetwork Designs for Improved Classification and Robust Meta-Learning Sudarshan Babu, Pedro Savarese, Michael Maire

* equal contribution † co-senior authorship

Service

Reviewing	CVPR, ICCV, ECCV, NeurIPS, ICLR, SIGGRAPH, RECOMB
Teaching	Teaching Assistant for Machine Learning, Prof. Greg Shakhnarovich, Fall 2020
Organization	Co-organizer: Computational Biology Workshop @ TTIC, Sep 2024; Annual TTI-C Student Workshop, Feb 2020; Computer Vision Reading Group, Oct 2020 – Feb 2024